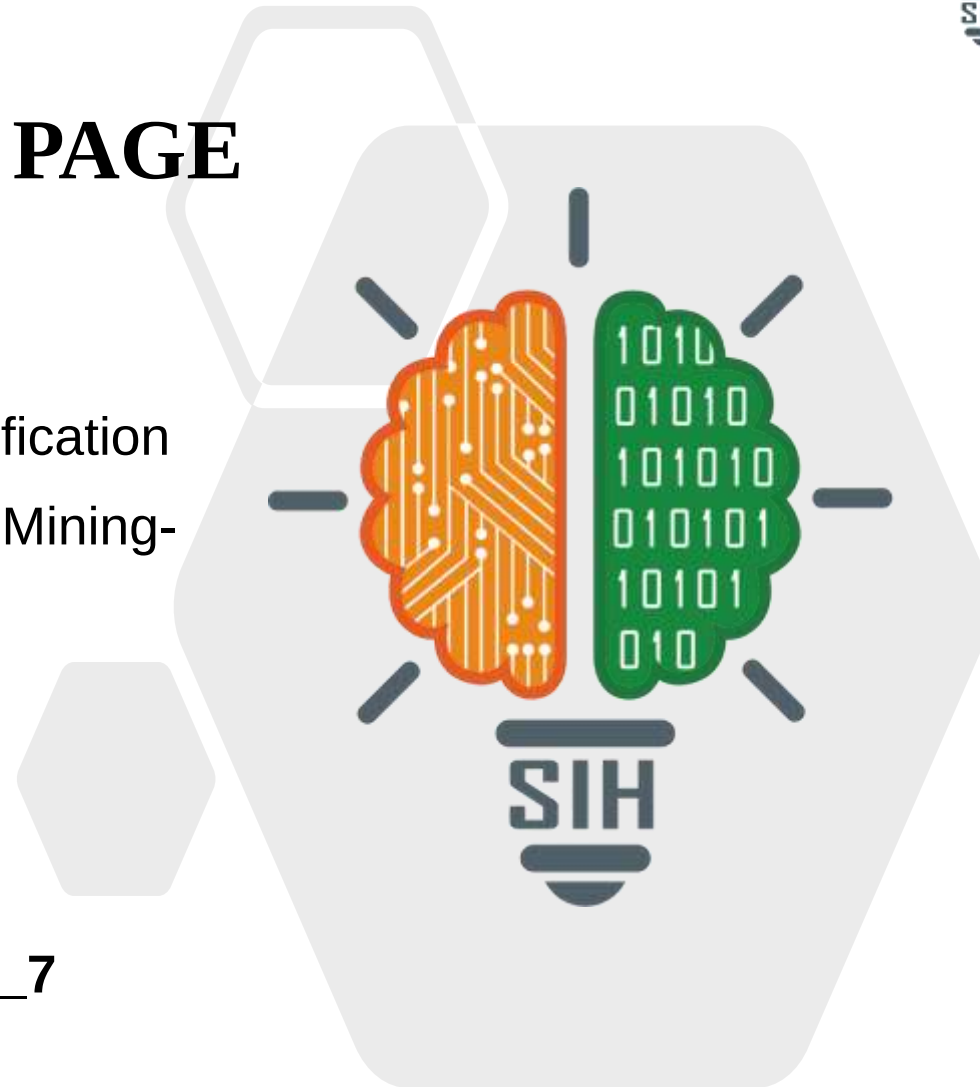


TITLE PAGE

- **Problem Statement ID** - SIH26040
- **Problem Statement Title** - Smart Water Purification and Quality Monitoring System for Rural and Mining-Affected Areas
- **Theme** - Clean & Green Technology
- **PS Category** - Hardware
- **Team ID** - 158819
- **Team Name (Registered on portal)** – Team_7



❖ Proposed Solution (Describe your Idea/Solution/Prototype)

Overview:

- Real-time sensing: Continuously monitors key water-quality parameters.
- Smart classification: Uses on-device ML to identify the type of contamination.
- Adaptive purification: Routes water through only the treatment stages it needs, making it suitable for variable mining-affected water.

Problems with conventional water treatment units:

- Fixed treatment pipeline regardless of contamination type
- No real-time visibility into water safety for the community
- Heavy-metal contamination from mining runoff often goes undetected
- High maintenance downtime with no early-warning of filter fouling

Innovation and uniqueness of the solution:

- On-device ML classifies contamination type, not just threshold breach
- Adaptive treatment routing activates only the stages needed per batch
- Electrochemical heavy-metal sensing (LMP91000 AFE) - rarely seen in low-cost systems
- Post-treatment verification with automatic reject/divert safety interlock
- Predictive filter-fouling alerts based on treatment efficiency trend

Prototype:

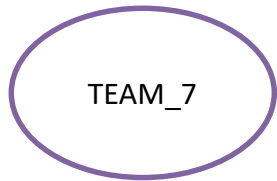
- Sensor cluster + control board housed in a sealed IP65 enclosure
- Multi-stage purification train: sediment -> carbon -> ion-exchange/electrocoagulation -> UV
- Solar + LiFePO₄ battery power unit for off-grid operation
- LoRa-based telemetry to a village-level dashboard, with GSM/SMS fallback

Concept explanation

Adaptive routing: sensor readings are classified on-device (TensorFlow Lite Micro) to identify contamination type before selecting a treatment path.

Electrochemical AFE sensing: a three-electrode setup with the LMP91000 front-end detects heavy metals (Fe, Mn, As) without lab-grade equipment.

Closed-loop verification: treated water is re-tested before entering storage; a failed batch is automatically diverted back for re-treatment.



TECHNICAL APPROACH



Hardware requirements:

- ESP32-S3 microcontroller unit
- pH, TDS, turbidity & ORP analog sensors
- DS18B20 digital temperature probe
- LMP91000 electrochemical AFE (heavy metals)
- ADS1115 16-bit external ADC
- Solenoid valves + MOSFET/relay drivers
- LoRa SX1278 module, GSM/SMS fallback
- Solar panel + LiFePO4 battery + charge controller

Software requirements

- Arduino / ESP-IDF firmware
- TensorFlow Lite Micro (edge classification)
- Proteus (circuit simulation)
- Custom dashboard / mobile app (BLE-MQTT bridge)

GitHub Repo : <https://github.com/Yuvaraaj-007/SiH-26040/tree/main>

WORKFLOW

Sensing & power	Processing	Communication	Actuation
Regulated 12V/5V/3.3V power distribution	Adaptive treatment stage selection	Real-time alerts & notifications	Usage logging & auto-refill trigger
Signal conditioning & buffering	On-device ML contamination classification	Mobile app data ingestion	Clean water storage & tank-level monitoring
pH, TDS, turbidity & ORP sensors	Sensor calibration & filtering	Cloud dashboard sync	Batch verification & reject/divert logic
LMP91000 electrochemical AFE (heavy metals)	ADS1115 16-bit ADC acquisition	GSM/SMS fallback alerts	Multi-stage purification: sediment, carbon, UV
Multi-source water intake	ESP32-S3 microcontroller unit	LoRa SX1278 module	12V pump/valve MOSFET driver

FEASIBILITY AND VIABILITY



ASPECT	FEASIBILITY	RISK/CHALLENGE	MITIGATION STRATEGY
Hardware and sensing	Low-cost analog sensors + LMP91000 AFE replace lab-grade heavy-metal testing equipment	Electrochemical sensor drift and fouling in silt-heavy mine runoff	Periodic auto-calibration cycle; pre-filter mesh to protect electrodes from silt
Computation and telemetry	ESP32-S3 runs TFLite Micro classification on-device; LoRa gives long range at low power	Limited connectivity in remote mining/rural zones; LoRa payload size constraints	GSM/SMS fallback for critical alerts; LoRa mesh relay between deployed units
Field and long-term viability	Solar + LiFePO4 removes grid dependency; multi-stage design adapts to seasonal contamination	Fluctuating solar input and battery degradation over multi-year field deployment	MPPT charge controller with margin sizing; predictive maintenance alerts for filter/battery health

IMPACT AND BENEFITS



- **RURAL COMMUNITIES** : Provides continuous access to verified safe drinking water without relying on periodic manual lab testing.
- **MINING-AFFECTED POPULATIONS** : Targets heavy-metal and acidic-drainage contamination specific to mining zones, which standard filters typically miss.
- **PUBLIC HEALTH IMPACT** : Reduces exposure to waterborne pathogens and heavy metals, lowering incidence of related illnesses in underserved areas.
- **ECONOMIC BENEFITS** : Off-grid solar operation and adaptive treatment reduce running cost and consumable replacement compared to fixed-pipeline purification units.
- **ENVIRONMENTAL BENEFITS** : Selective, on-demand treatment stage activation reduces unnecessary chemical/energy consumption versus always-on treatment trains.

Water Quality Classification Using Machine Learning Algorithms – Elsevier

<https://www.sciencedirect.com/science/article/pii/S2214714422003646>

Real Time Monitoring of Water Quality for Rural Areas – IEEE

<https://ieeexplore.ieee.org/document/10165824/>

Resource Utilization of Acid Mine Drainage (AMD): A Review – MDPI

<https://www.mdpi.com/2073-4441/14/15/2385>

Electrochemical Sensors for Heavy Metal Ion Detection in Aqueous Medium – ACS

<https://pubs.acs.org/acsodf/article/9/24/25493/410316/Electrochemical-Sensors-for-Heavy-Metal-Ion>

Recent Trends in Electrochemical Methods for Real-Time Detection of Heavy Metals – Elsevier

<https://www.sciencedirect.com/science/article/pii/S2451910325001085>

Membrane Fouling Prediction Using Machine Learning – Elsevier

<https://www.sciencedirect.com/science/article/pii/S0376738822005622>

GitHub Repo : <https://github.com/Yuvaraaj-007/SiH-26040/tree/main>